Week 9

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Explain the role of probability in statistics. (CLO 4)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe how a randomization distribution is generated assuming H_0 is true. (CLO 5, CLO 8) **2.** Identify the center of a randomization distribution as the null value. (CLO 5) **3.** Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CLO 5) **4.** Compute a p-value from a randomization distribution as the proportion of simulated statistics at least as extreme as the observed statistic. (CLO 3, CLO 8) **5.** Determine the correct tail(s) for a p-value from the alternative hypothesis. (CLO 8) **6.** Interpret a p-value in context. (CLO 4, CLO 8)	e-Text: 4.2 Measuring Evidence with P-Values Interactive Lecture: 4.2 Measuring Evidence with P-Values Slides: 4.2 Measuring Evidence with P-Values	Participation: P-Values – StatKey: dolphin study (16 flips); count how many simulated statistics are as extreme as observed; Your Turn: Tea and Immunity in StatKey (re-randomize tea vs. coffee, find the p-value)	Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM) Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 4.1–4.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Respondus
7. Relate the size of a p-value to the strength of evidence against H_0. (CLO 8) **8.** Compare a p-value to the significance level α to decide whether to reject H_0. (CLO 8) **9.** Write a two-step conclusion in context that avoids saying "accept H_0". (CLO 8)	e-Text: 4.3 Determining Statistical Significance Interactive Lecture: 4.3 Determining Statistical Significance Slides: 4.3 Determining Statistical Significance (includes "What is a p-value?" video)	Participation: Determining Statistical Significance – resveratrol examples (p = 0.04, 0.12, 0.0003): strength of evidence, decision at α, conclusion in context; Your Turn: Decide and Conclude	Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM) Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 4.1–4.3 (all objectives) Test 2 (Fri Oct 30 – Sun Nov 1, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Oct 29) (all objectives)	WileyPLUS StatKey Blackboard Zoom Respondus